

A New Qubits Mapping Mechanism for Multi-programming Quantum Computing

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ABSTRACT

For a specific quantum chip, multi-programming helps to improve the overall throughput and resource utilization. However, previous solutions for mapping multiple programs often lead to resource under-utilization, high error rate, and low fidelity. In this paper, we propose a new approach to map concurrent quantum programs. Our approach has three critical components. The first one is the Community Detection Assisted Partition (CDAP) algorithm, which partitions physical qubits for concurrent quantum programs by considering physical typology and the error rate, avoiding the waste of robust resources. The second one is the X-SWAP scheme that enables inter-program SWAPs and prioritizes SWAPs associated with critical gates to reduce the SWAP overheads. Finally, we propose a compilation task scheduler, which dynamically selects concurrent quantum programs to be compiled and executed together based on estimated fidelity for the best practice. We evaluate our work on publicly available quantum computer IBMQ16 and a simulated quantum chip IBMQ50. Our work outperforms the state-of-the-art work for multi-programming on fidelity and compilation overheads by 12.0% and 11.6%, respectively.

1 INTRODUCTION

Quantum computer is stepping into our view. However, modern quantum chips belong to Noisy Intermediate-Scale Quantum (NISQ) category [1] - the qubits and the links between them are with variational reliability and are easily disturbed. The emergence of quantum cloud services enables users to access the quantum computers easily, but it also brings new challenges. As NISQ computers exhibit low fidelity, only programs with a few qubits can be executed reliably. Thus, NISQ computers tend to under-utilize their resources. Multi-programming can be an effective way. Although mapping multiple quantum programs onto a specific quantum chip improves the throughput, the activity of a program can negatively affect the reliability of co-located programs because of (i) limited number of qubits with high fidelity; (ii) cross-talk noise caused by simultaneously executed quantum gates; and (iii) long SWAP paths.

In this paper, we propose solutions that improve the throughput and utilization of NISQ machines while reducing the negative impacts on the reliability of multi-programming NISQ computers. We find the previous qubits mapping policies have several shortcomings when handling multi-programming cases. (1) They often divide a large area of robust on-chip qubits into many small-scale segments that cannot be mapped onto for other programs. On average, over 20% of the robust qubits are wasted during the initial mapping. (2) When a specific quantum chip is partitioned for mapping multiple quantum programs, post-compilation SWAP operations for each quantum program can be more, leading to an unpredictable impact for fidelity and reliability. For instance, additional SWAPs can be involved when two quantum programs with tens of CNOT gates are compiled together. (3) There is no ideal approach to select concurrent quantum programs for multi-programming on a specific quantum chip, leading to fidelity degradation and qubit resource under-utilization in many cases.

Towards this end, we design a new qubits mapping mechanism, which has two key features. First, it partitions the physical qubits for concurrent quantum programs leveraging community detection techniques, avoiding the waste caused by the typology-unaware algorithms. It also provides a better initial-mapping, which reduces the SWAP overheads. Second, it is the First work that enables the inter-program SWAPs to solve the mapping problem in multi-programming cases that reduces the overall SWAP overheads. Our approach works well in practice. The experimental results show that our approach outperforms the latest solution [2] by 12.0% on fidelity and 11.6% on compilation overheads. Furthermore, we design a scheduler for selecting compilation tasks for the best practice of multi-programming, avoiding the performance degradation caused by randomly selected workloads. The scheduler improves the throughput by 42.9% and enhances the fidelity by 5.0% over randomly selected workloads in our experiments, on average.

2 THE ART OF OUR DESIGN

2.1 A new qubits mapping policy - CDAP

In this paper, we propose a new technology - Community Detection Assisted Partitioning (CDAP) - to construct a hierarchy tree consisted of qubits for searching the robust qubits that are tightly connected for initial allocation. CDAP creates a hierarchy tree according to the coupling map and calibration data obtained from IBMQ API [3]. In the hierarchy tree, a leaf node denotes a specific physical qubit; an internal node represents the union of its sub-nodes. Each node in the hierarchy tree is a candidate region for initial allocation. CDAP then iterates the hierarchy tree from bottom to top to find available regions for initial allocation. The quantum circuits are allocated by greedy policy to corresponding regions. The hierarchy tree is a profile of a quantum chip, which helps to locate reliable qubit resources on the quantum computer. The hierarchy tree construction algorithm is based on FN community detection algorithm, which clusters the physical qubits on a specific quantum chip into communities. Qubits in a community have reliable and close interconnections. In contrast, the links between

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PACT '20, October 3–7, 2020, Virtual Event, GA, USA
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ACM ISBN 978-1-4503-8075-1/20/10.
https://doi.org/10.1145/3410463.3414659

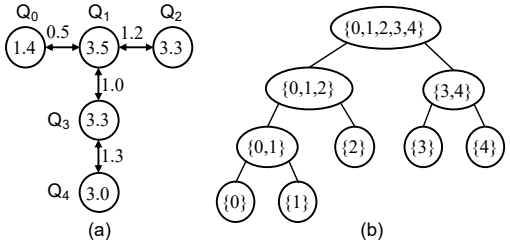


Figure 1: (a) Architecture of IBM Q London. The value in a node represents the readout error rate of the qubit, and the value on a link means the error rate of the CNOT operation. (b) The dendrogram, in which the values denote the index of the physical qubits.

communities have relatively low reliability.

We explain why the hierarchy tree helps to select the initial allocation with an example in Figure 1. (i) Q_0 and Q_1 are firstly merged due to the link between them is with the lowest error rate. (ii) Then, Q_2 instead of Q_3 is merged into the community {0, 1}, though the link Q_1-Q_3 has lower CNOT error rate than Q_1-Q_2. This is because we tend to merge more inter-connected nodes into one community, avoiding the waste of robust physical qubits. Likewise, Q_3 and Q_4 are merged. (iii) Finally, all qubits are merged as the root of the hierarchy tree as illustrated in Figure 1-(b). Our approach avoids the waste of robust resources caused by the typology-unaware greedy algorithm, and supports more quantum programs to be mapped on a specific quantum chip.

2.2 The design of inter-program SWAP - X-SWAP

Multi-programming brings new challenges for mapping transition. In this paper, we design the X-SWAP, including both inter and intra-program SWAP operations. In practice, the inter-program SWAP operation can be enabled when quantum programs are close to each other. The cost of inter-program SWAPs can be less than the cost in the cases where intra-program SWAPs are merely used.

In our study, we find inter-program SWAPs take shortcuts. For instance, Figure 2-(a) shows two quantum programs are co-located (mapped) on a quantum chip with nine qubits. q_1 and q_5 are not mapped physically adjacent; SWAPs are required to satisfy their constraint to make CNOT q_1, q_5 executable. As illustrated in Figure 2-(b), an inter-program SWAP, i.e., $\{q_1, q_9\}$, takes only one step (swap operation) to move q_1 and q_5 adjacent. In contrast, to achieve the same goal, previous intra-program scheme has to introduce three SWAPs, i.e., $\{q_1, q_2\}$, $\{q_1, q_3\}$, $\{q_1, q_4\}$. Briefly, enabling inter-program SWAPs could result in fewer SWAPs in the cases where multiple quantum programs are mapped as neighbors on a specific quantum chip, therefore reducing the SWAP overheads and benefiting the overall fidelity.

2.3 The design of the compilation task scheduler

Although there are some mapping mechanisms for concurrent quantum programs, selecting appropriate quantum programs to form a combination for the multi-programming workload is still a challenging job. The workloads are now selected manually, which may cause the following defects. (1) Qubits on a specific quantum chip are under-utilized. (2) The program combinations formed by randomly selected programs may lead to a significant reduction in

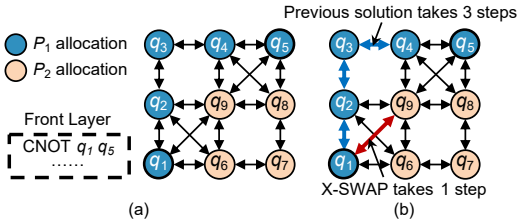


Figure 2: (a) P_1 and P_2 are mapped on a quantum chip with 9 qubits. The next gate to be solved is CNOT that involves q_1 and q_5. (b) X-SWAP scheme takes shortcuts to satisfy the constraint of CNOT q_1, q_5.

fidelity. (3) The results verification mechanism must be introduced to ensure the fidelity, which brings additional system modification overheads. To this end, we propose a design for the compilation task scheduler to select appropriate concurrent quantum programs for multi-programming. Our design focuses on selecting optimal quantum program combinations, maximizing the fidelity and resource utilization of the quantum chip. For each task in the scheduling queue, the scheduler checks whether other quantum programs in the queue can bring allowable fidelity reduction (e.g., the maximum reduction defined by the users) when they are co-located on the quantum chip with the current task. If so, they are mapped to the target quantum computer simultaneously for multi-programming. Otherwise, the task will be executed independently.

3 EVALUATION

The baseline is the policy proposed in [2], which generates initial mapping for concurrent quantum programs with FRP strategy and generates mapping transition with the enhanced noise-aware SABRE. With our design, CDAP creates a reliable and close interconnected initial mapping; X-SWAP reduces the compilation overheads. CDAP and X-SWAP work together to benefit the performance - reducing the number of post-compilation gates by 11.6% compared with baseline, and 8.6% compared with SABRE [4]. The circuit depth is reduced by 16.0% and 10.3% compared with baseline and SABRE, respectively. The compilation task scheduler trades off between throughput and fidelity. It outperforms randomly selected workloads by 5.0% on fidelity and improves the throughput of the quantum computer by 42.9%. Moreover, our work exhibits scalability. It helps to improve the fidelity of the 2-program workloads on IBMQ16. For 4-program workloads on a quantum chip with 50 qubits, it reduces the compilation overheads by 11.6%.

ACKNOWLEDGMENTS

This project is supported by the National Key Research and Development Program of China under Grant No. 2017YFB1001602 and the NSF of China under Grant No. 61502452. Xinglei Dou is a student member in Sys-Inventor Lab supervised by Lei Liu.

REFERENCES

[1] J. Preskill, "Quantum Computing in the NISQ era and beyond," Quantum, vol. 2, p. 79, 2018.
[2] P. Das, et al, "A Case for Multi-Programming Quantum Computers," in Micro, 2019.
[3] "IBM Quantum Experience," https://quantum-computing.ibm.com/, Accessed April 2, 2020.
[4] G. Li, et al, "Tackling the qubit mapping problem for NISQ-era quantum devices," in ASPLOS, 2019.